MATH 6660 — Homework 3

Beam equation (cantilever)

Fall 2026

Relation to the lectures. Lectures 10–11 derive the weak form of the beam equation $u^{(iv)} = f$ and Lecture 12 carries CG(3) through to the assembled system — but for a beam *clamped at both ends*, $u(0) = u(1) = u'(0) = u'(1) = 0$. This homework is the same beam with the right end *free*. Only one thing changes: which of the four boundary terms are killed by a condition on v and which are killed by a condition on u. The machinery — two integrations by parts, the space built from $\int[v^2 + (v')^2 + (v'')^2] < \infty$, Hermite cubics, Galerkin orthogonality, the duality argument — is used exactly as the lectures state it.

Problem 1

Problem. Recast the cantilever beam problem as a variational problem. Clearly state the test and trial function spaces.

The strong problem is

$$\begin{cases} u^{(iv)} = f(x), & x \in (0,1), \\ u(0) = 0, \quad u'(0) = 0, & \text{(clamped end)} \\ u''(1) = 0, \quad u'''(1) = 0. & \text{(free end)} \end{cases} \tag{1}$$

Step 1 — multiply by v and integrate by parts twice [Lecture 11: same two steps, $\int v\,u^{(iv)} \to -\int v'u''' \to \int v''u''$]

- Multiply (1) by a test function v and integrate over $(0,1)$:

$$\int_0^1 v\,u^{(iv)}\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x. \tag{2}$$

- First integration by parts:

$$\underbrace{\Big[v\,u'''\Big]_0^1}_{\text{bracket A}} - \int_0^1 v'\,u'''\,\mathrm{d}x \;=\; \int_0^1 fv\,\mathrm{d}x. \tag{3}$$

- Second integration by parts, applied to $-\int v'u'''$:

$$\text{bracket A}\ \underbrace{-\Big[v'\,u''\Big]_0^1}_{\text{bracket B}} + \int_0^1 v''\,u''\,\mathrm{d}x \;=\; \int_0^1 fv\,\mathrm{d}x. \tag{4}$$

Step 2 — kill the two brackets [Lecture 6: Dirichlet = essential, built into V; Neumann = natural, no constraint on the space]

Write the brackets out. Each has an $x = 1$ half and an $x = 0$ half, and the two halves die for *different reasons*:

$$\mathrm{A} = \underbrace{v(1)\,u'''(1)}_{x=1} - \underbrace{v(0)\,u'''(0)}_{x=0}, \qquad \mathrm{B} = -\underbrace{v'(1)\,u''(1)}_{x=1} + \underbrace{v'(0)\,u''(0)}_{x=0}.$$

term	dies because	type	costs a condition on
$v(1)\,u'''(1)$	$u'''(1)=0$ is *given* in (1)	natural	nothing
$v'(1)\,u''(1)$	$u''(1)=0$ is *given* in (1)	natural	nothing
$v(0)\,u'''(0)$	we *impose* $v(0)=0$	essential	v
$v'(0)\,u''(0)$	we *impose* $v'(0)=0$	essential	v

- At the **free end** $x=1$ nothing is asked of v: the two boundary data $u''(1)=u'''(1)=0$ are already zero, so both $x=1$ halves vanish on their own. This is Lecture 6's "natural, weakly enforced, adds no constraint".
- At the **clamped end** $x=0$ the data $u'''(0)$ and $u''(0)$ are *not* known — they are the unknown reaction shear and reaction moment. The only way to remove them is to demand $v(0)=v'(0)=0$. This is Lecture 6's "essential, strongly enforced, built into V".
- Both brackets are now zero, and (4) collapses to

$$\boxed{\int_0^1 u''\,v''\,\mathrm{d}x=\int_0^1 f\,v\,\mathrm{d}x} \tag{5}$$

Step 3 — the space [Lecture 11, eq. (1.170): same recipe, only the BC list is shorter]

- (5) asks for $\int(u'')^2$ and $\int(v'')^2$ to be finite, so the space is built on two derivatives — exactly as in Lecture 11:

$$\boxed{V=\Big\{v:\ \int_0^1\big[v^2+(v')^2+(v'')^2\big]\,\mathrm{d}x<\infty,\quad v(0)=0,\ v'(0)=0\Big\}} \tag{6}$$

- Only the *two* essential conditions appear in (6). The lecture's clamped–clamped version lists four because that beam has two clamped ends; the cantilever has one. The free-end conditions $u''(1)=u'''(1)=0$ are **not** written into V — they are already built into (5), and a solution of (5) satisfies them whether or not anybody asks.
- As in Lecture 11, functions in V are $C^1[0,1]$: a corner in v puts a jump in v', hence a delta in v'', hence $\int(v'')^2=\infty$.

Step 4 — trial and test spaces [Lecture 5: trial = test exactly when the essential BCs are homogeneous]

- **Trial space** $=V$. The trial function is the unknown u; it must satisfy the essential conditions $u(0)=u'(0)=0$, and it must have finite bending energy $\int(u'')^2<\infty$. That is (6).
- **Test space** $=V$. Step 2 forced $v(0)=v'(0)=0$ and (5) forced $\int(v'')^2<\infty$. Same two requirements, same set.
- They coincide because the essential BCs here are *homogeneous* (0, not a prescribed settlement). Lecture 5 is the case where they part: with $u(0)=\alpha\neq 0$ the trial set is not even a vector space, and one uses V for u and V_0 for v. That does not arise here.

(Variational problem) Find $u \in V$ such that

$$\int_0^1 u''v''\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x \qquad \text{for all } v \in V,$$

with V given by (6), and trial space = test space = V.

Mechanics reading. $\int u''v''$ is internal virtual work (curvature $\times$ moment, with $EI \equiv 1$); $\int fv$ is external virtual work. The statement is virtual work. $v(0) = v'(0) = 0$ says a virtual displacement may not move or rotate the built-in end — you cannot do virtual work on a support. Nothing is asked at the free tip because a free tip carries no reaction.

Problem 2

Problem. Interpret the variational problem as an energy minimization problem, clearly stating the energy functional. Prove that the variational problem and the energy minimization problem are equivalent.

Step 1 — the energy functional [Lecture 11, eq. (1.171): $F(w) = \frac{1}{2}\int (w'')^2 - \int fw$ for this very beam]

- Lecture 3 built F for the bar as *internal energy − load potential*, $F(w) = \frac{1}{2}\int a(w')^2 - \int fw$ (eq. 1.29). Lecture 11 writes the beam version, one derivative higher:

$$\boxed{F(w) = \underbrace{\frac{1}{2}\int_0^1 (w'')^2 \,\mathrm{d}x}_{\text{bending strain energy}} - \underbrace{\int_0^1 f\, w \,\mathrm{d}x}_{\text{load potential}}} \tag{7}$$

- w'' is the curvature, so with $EI \equiv 1$ the first term is $\frac{1}{2}\int EI\,\kappa^2$ — the bending strain energy, as Lecture 11 states. F is the total *potential* energy $\Pi = U - W$.

(Minimization problem) Find $u \in V$ such that $F(u) \le F(w)$ for all $w \in V$, with V from (6) and F from (7).

This is the **principle of minimum total potential energy**: of all admissible deflected shapes of the cantilever — all shapes that keep the built-in end built in — the one that actually occurs is the one with the least potential energy.

Step 2 — a remark used by both directions

- V in (6) is a **vector space**: its two constraints $v(0) = 0$, $v'(0) = 0$ are homogeneous, so sums and scalar multiples of members stay inside.
- Hence: if $u, w \in V$ then $v := w - u \in V$; and if $u, v \in V$ then $u + \varepsilon v \in V$ for every real ε. Both directions below need this, and it is exactly what fails in Lecture 5's non-homogeneous case.

Step 3 — (Var. Prob.) $\Longrightarrow$ (Min. Prob.) [Lecture 11, eq. (1.172)–(1.173): expand $F(u+v)$, the cross terms are the variational problem, so they cancel]

- Let $u \in V$ solve the variational problem (5), and take any $w \in V$. Write $w = u + v$ with $v = w - u \in V$ (Step 2).
- Expand, using $(u'' + v'')^2 = (u'')^2 + 2u''v'' + (v'')^2$:

$$\begin{aligned} F(w) = F(u+v) &= \frac{1}{2}\int_0^1 (u'')^2 \,\mathrm{d}x + \int_0^1 u''v'' \,\mathrm{d}x + \frac{1}{2}\int_0^1 (v'')^2 \,\mathrm{d}x - \int_0^1 fu \,\mathrm{d}x - \int_0^1 fv \,\mathrm{d}x \\ &= \underbrace{\frac{1}{2}\int_0^1 (u'')^2 \,\mathrm{d}x - \int_0^1 fu \,\mathrm{d}x}_{=F(u)} + \frac{1}{2}\int_0^1 (v'')^2 \,\mathrm{d}x + \underbrace{\int_0^1 u''v'' \,\mathrm{d}x - \int_0^1 fv \,\mathrm{d}x}_{=0 \text{ by (5), with this } v}. \end{aligned} \tag{8}$$

- The last bracket is the variational problem read with test function v, so it vanishes. What is left is

$$\boxed{F(w) = F(u) + \frac{1}{2}\int_0^1 (v'')^2 \,\mathrm{d}x} \tag{9}$$

- The remaining integral is a square, hence ≥ 0. Therefore $F(w) \geq F(u)$ for all $w \in V$, which is the minimization problem.

Reading (9). Perturb the true shape u by any admissible v. The first-order change is exactly the variational problem and vanishes *because u is in equilibrium.* What survives is second order: $\frac{1}{2}\int (v'')^2$, the bending energy stored by the error itself. Equilibrium sits at the bottom of an energy well, and any wrong shape costs precisely the strain energy of being wrong.

Step 4 — (Min. Prob.) $\Longrightarrow$ (Var. Prob.) [Lecture 3, eq. (1.36): set $g(\varepsilon) = F(u+\varepsilon v)$, note g is a quadratic in ε minimised at $\varepsilon = 0$, so $g'(0) = 0$]

- Let $u \in V$ satisfy $F(u) \leq F(w)$ for all $w \in V$. Fix an arbitrary $v \in V$ and define the one-variable function

$$g(\varepsilon) = F(u + \varepsilon v), \qquad \varepsilon \in \mathbb{R}. \tag{10}$$

- By Step 2, $u + \varepsilon v \in V$ for every ε, so $g(\varepsilon) \geq F(u) = g(0)$ for every ε: g **has a global minimum at** $\varepsilon = 0$.
- Expand g exactly as in Step 3, but keeping ε:

$$g(\varepsilon) = \underbrace{\frac{1}{2}\int_0^1 (u'')^2 \,\mathrm{d}x - \int_0^1 fu \,\mathrm{d}x}_{\text{const.}} + \varepsilon \underbrace{\left[\int_0^1 u''v'' \,\mathrm{d}x - \int_0^1 fv \,\mathrm{d}x\right]}_{=:\,c_1} + \frac{\varepsilon^2}{2}\underbrace{\int_0^1 (v'')^2 \,\mathrm{d}x}_{=:\,c_2 \geq 0}. \tag{11}$$

  So g is a **quadratic polynomial in** ε, with $c_2 \geq 0$.
- Differentiate:

$$g'(\varepsilon) = \int_0^1 u''v'' \,\mathrm{d}x - \int_0^1 fv \,\mathrm{d}x + \varepsilon \int_0^1 (v'')^2 \,\mathrm{d}x \;=\; c_1 + \varepsilon\, c_2. \tag{12}$$

- g is differentiable and has an interior global minimum at $\varepsilon = 0$, hence $g'(0) = 0$, i.e.

$$c_1 = \int_0^1 u''v'' \,\mathrm{d}x - \int_0^1 fv \,\mathrm{d}x = 0 \qquad \Longrightarrow \qquad \int_0^1 u''v'' \,\mathrm{d}x = \int_0^1 fv \,\mathrm{d}x. \tag{13}$$

- $v \in V$ was arbitrary, so this holds for all $v \in V$: u solves the variational problem (5). ■

Why $g'(0) = 0$ really follows. If $c_2 > 0$ the parabola opens upward and its unique minimiser is $\varepsilon = -c_1/c_2$; that must be 0, so $c_1 = 0$. If $c_2 = 0$ then g is the straight line $g(0) + c_1\varepsilon$, and a straight line is minimised at $\varepsilon = 0$ over all of $\mathbb{R}$ only when $c_1 = 0$. Either way $c_1 = 0$. (In fact $c_2 = 0$ forces $v \equiv 0$ by Step 5, so only the first case really occurs.)

Step 5 — the minimiser is unique [Lecture 11, Note: $F(w) = F(u) \Rightarrow v'' \equiv 0 \Rightarrow v \equiv 0$ by the BCs]

- From (9), $F(w) = F(u)$ forces $\int_0^1 (v'')^2 \,\mathrm{d}x = 0$, hence $v'' \equiv 0$.

- $v'' \equiv 0$ only says v is a straight line, $v = \alpha + \beta x$. It is the *clamped-end* conditions in (6) that kill it:

$$v(0) = 0 \;\Rightarrow\; \alpha = 0, \qquad v'(0) = 0 \;\Rightarrow\; \beta = 0, \qquad \text{so } v \equiv 0, \quad w = u.$$

- So $F(w) > F(u)$ for every $w \neq u$: the minimiser is unique, and by the equivalence so is the solution of the variational problem.

The cantilever's two clamped conditions are exactly enough. They are what remove the rigid-body translation α and rotation β — a beam free at *both* ends would keep them, F would not have a unique minimiser, and K would be singular.

Conclusion. With F as in (7): u solves $\displaystyle\int_0^1 u''v''\,\mathrm{d}x = \int_0^1 fv\,\mathrm{d}x \quad \forall v \in V \iff F(u) \leq F(w) \quad \forall w \in V$, and the solution of either is unique.

Problem 3

Problem. Develop a CG(3) finite element method for this problem.

Step 1 — why cubics, and why Hermite [Lecture 11: $V_h \subseteq V$ needs value *and* derivative matched at each node — 4 conditions per element, so a cubic]

- The method must be **conforming**, $V_h \subset V$. The only smoothness V demands is $\int_0^1 (v'')^2 \,\mathrm{d}x < \infty$ from (6). Test what that permits at a node, one derivative at a time:

at a node	then v'' contains	$\int_0^1 (v'')^2 \,\mathrm{d}x$	verdict
v jumps	a derivative of a delta	∞	forbidden
v continuous, v' jumps	a delta	∞	forbidden
v, v' continuous, v'' jumps	a finite jump	finite	**allowed**

- So the requirement is exactly $v \in C^1$: v **and** v' **must be continuous across every node;** v'' **need not be.** "No kink" means no kink *in* v — a kink in the graph of v is a jump in the slope v', which is the second row of the table.
- The third row is the useful half of the statement: because v'' is *allowed* to jump, the elements only have to be glued to order 1, not order 2. Continuity of v and of v' is 2 conditions at each end of an element, so 4 per element — a polynomial with 4 free coefficients, i.e. a **cubic**. Hence CG(3).
- A piecewise quadratic has only 3 coefficients and cannot meet all four, and a piecewise quintic would meet them with 2 coefficients to spare — **cubic is the smallest degree that works**.

Step 2 — mesh and which degrees of freedom survive [Lecture 6: the clamped end removes a DOF; the free end keeps it — "the +1 is because the last node is unknown"]

- Mesh, in the lecture's convention (Lecture 3–4): M *interior* nodes, hence $M+1$ elements,

$$0 = x_0 < x_1 < \cdots < x_M < x_{M+1} = 1, \qquad \underbrace{h_j = x_j - x_{j-1}}_{\text{element length}}, \quad j = 1, \ldots, M+1. \tag{14}$$

  Uniform mesh: $h_j = h$ for all j, so

$$\boxed{h = \frac{1}{M+1}} \qquad \text{and} \qquad x_j = jh. \tag{15}$$

  h **is the element length, not a count**; M counts *interior* nodes, so the number of elements is $M+1$.
- Each node carries **two** degrees of freedom: a deflection ξ_j and a slope η_j.
- **Clamped end** $x_0 = 0$**:** $v(0) = v'(0) = 0$ is essential, so both DOFs at node 0 are deleted.
- **Free end** $x_{M+1} = 1$**:** nothing is imposed (Problem 1, Step 2), so *both* DOFs at node $M+1$ are kept. This is Lecture 6's Neumann end node — "the +1 is because the last node is unknown".
- Hence

$$\boxed{\begin{aligned} V_h &= \big\{ v \in V : \ v|_{[x_{j-1},x_j]} = a_j + b_j x + c_j x^2 + d_j x^3, \ \ j = 1, \ldots, M+1 \big\} \\ &= \operatorname{span}\big\{ \phi_1^{(1)}, \ldots, \phi_{M+1}^{(1)}, \phi_1^{(2)}, \ldots, \phi_{M+1}^{(2)} \big\}, \qquad \dim V_h = 2(M+1) \end{aligned}} \tag{16}$$

Step 3 — the basis [Lecture 11, eqs. (1.175)–(1.176) and its nodal table]

Node j carries two DOFs, so it needs two basis functions. Lecture 11 fixes them by the nodal table — read a column as "what this function and its slope do at that node":

	x_{j-1}	x_j	x_{j+1}
$\phi_j^{(1)}$	0	1	0
$(\phi_j^{(1)})'$	0	0	0
$\phi_j^{(2)}$	0	0	0
$(\phi_j^{(2)})'$	0	1	0

- Superscripts (1) and (2) are labels, not powers: $\phi_j^{(1)}$ deflects node j by 1 without tilting it, $\phi_j^{(2)}$ gives node j slope 1 without moving it. ("Slope" and "rotation" are the same quantity, U'; this note says *slope* throughout.) Both are zero at every other node, so each lives on only the two elements at node j.
- **Why the table is imposed.** Put $U = \sum \xi_j \phi_j^{(1)} + \sum \eta_j \phi_j^{(2)}$ through it. At node k every term dies except one, leaving

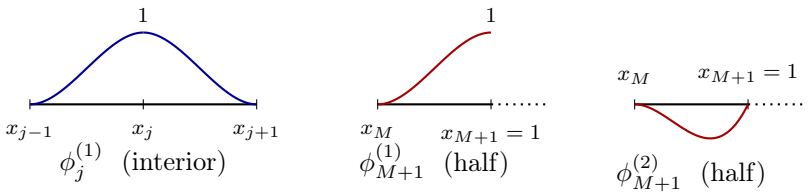
$$\boxed{U(x_k) = \xi_k, \qquad U'(x_k) = \eta_k} \tag{17}$$

  so ξ_k *is* the deflection at node k and η_k *is* the slope there. The four zero rows are what keep the two DOFs from mixing; without them ξ_k would be some combination of deflection and slope and (17) would fail.
- The explicit shapes (Lecture 11, uniform mesh):

$$\phi_j^{(1)} = \begin{cases} 3\Big(\dfrac{x - x_{j-1}}{h}\Big)^2 - 2\Big(\dfrac{x - x_{j-1}}{h}\Big)^3, & x_{j-1} \le x \le x_j, \\ 3\Big(\dfrac{x - x_{j+1}}{-h}\Big)^2 - 2\Big(\dfrac{x - x_{j+1}}{-h}\Big)^3, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise}, \end{cases} \tag{18}$$

$$\phi_j^{(2)} = \begin{cases} -\dfrac{1}{h}(x - x_{j-1})^2 + \dfrac{1}{h^2}(x - x_{j-1})^3, & x_{j-1} \le x \le x_j, \\ \dfrac{1}{h}(x - x_{j+1})^2 + \dfrac{1}{h^2}(x - x_{j+1})^3, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise}. \end{cases} \tag{19}$$

- **At the tip $j = M + 1$ only the first branch exists** — there is no element to its right. $\phi_{M+1}^{(1)}, \phi_{M+1}^{(2)}$ are *half* shapes on $[x_M, 1]$: the Hermite version of Lecture 5's half-hat, and the reason the last row of the matrix changes in Step 5.



Step 3b — the same thing element by element [Lecture 11 mechanics note: "$\phi_j^{(1)}$ and $\phi_j^{(2)}$ are the cubic Hermite shape functions N_1, N_2, N_3, N_4"]

In a structural code the trial function is written per *element*, not per node. It is the same function. On element $e = [x_{e-1}, x_e]$, with $t = x - x_{e-1}$,

$$U\big|_e = N_1 w_{e-1} + N_2 \theta_{e-1} + N_3 w_e + N_4 \theta_e, \qquad \begin{aligned} N_1 &= 1 - 3\tfrac{t^2}{h^2} + 2\tfrac{t^3}{h^3}, & N_2 &= t - 2\tfrac{t^2}{h} + \tfrac{t^3}{h^2}, \\ N_3 &= 3\tfrac{t^2}{h^2} - 2\tfrac{t^3}{h^3}, & N_4 &= -\tfrac{t^2}{h} + \tfrac{t^3}{h^2}. \end{aligned}$$

Comparing with (18)–(19): $\phi_j^{(1)}$ is N_3 on the element to its left and N_1 on the element to its right; $\phi_j^{(2)}$ is N_4 then N_2. One global basis function = two element shape functions glued at the node, and the coefficients are the same numbers, $w_e = \xi_e$, $\theta_e = \eta_e$ — which is why assembling element matrices into global slots (Step 5) works.

Step 4 — the CG(3) method [Lecture 11 box; Lecture 12, eq. (1.181): trial U, test = each basis function]

- **Trial function:**

$$U(x) = \sum_{j=1}^{M+1} \xi_j \, \phi_j^{(1)}(x) + \sum_{j=1}^{M+1} \eta_j \, \phi_j^{(2)}(x), \qquad \xi_j = U(x_j), \quad \eta_j = U'(x_j). \tag{20}$$

- **Test functions:** $v = \phi_k^{(1)}, \phi_k^{(2)}$, $k = 1, \ldots, M+1$.

(CG(3) FEM) Find $U \in V_h$ such that $\displaystyle\int_0^1 U''v'' \,\mathrm{d}x = \int_0^1 fv \,\mathrm{d}x$ for all $v \in V_h$.

Step 5 — the linear system [Lecture 12, eqs. (1.189)–(1.194): the A, B, C blocks]

- Substituting (20) and testing against each basis function gives $2(M+1)$ equations in the $2(M+1)$ unknowns ξ, η. Sorting them by *type* of DOF gives Lecture 12's block form

$$\underbrace{\left[\begin{array}{c|c} \boldsymbol{A} & \boldsymbol{C} \\ \hline \boldsymbol{C}^T & \boldsymbol{B} \end{array}\right]}_{\boldsymbol{S} \in \mathbb{R}^{2(M+1)\times 2(M+1)}} \left[\begin{array}{c} \boldsymbol{\xi} \\ \hline \boldsymbol{\eta} \end{array}\right] = \left[\begin{array}{c} \boldsymbol{b}^{(1)} \\ \hline \boldsymbol{b}^{(2)} \end{array}\right], \tag{21}$$

$$A_{kj} = \int_0^1 (\phi_j^{(1)})''(\phi_k^{(1)})'' \,\mathrm{d}x, \quad B_{kj} = \int_0^1 (\phi_j^{(2)})''(\phi_k^{(2)})'' \,\mathrm{d}x, \quad C_{kj} = \int_0^1 (\phi_j^{(2)})''(\phi_k^{(1)})'' \,\mathrm{d}x.$$

- On a uniform mesh these are Lecture 12's matrices — **except in the last row and column**, where node $M+1$ collects only *one* element instead of two:

$$\boldsymbol{A} = \frac{12}{h^3}\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & 2 & -1 \\ & & -1 & \boxed{1} \end{bmatrix}, \quad \boldsymbol{B} = \frac{2}{h}\begin{bmatrix} 4 & 1 & & \\ 1 & 4 & \ddots & \\ & \ddots & 4 & 1 \\ & & 1 & \boxed{2} \end{bmatrix}, \quad \boldsymbol{C} = \frac{6}{h^2}\begin{bmatrix} 0 & 1 & & \\ -1 & 0 & \ddots & \\ & \ddots & 0 & 1 \\ & & -1 & \boxed{-1} \end{bmatrix}. \tag{22}$$

- All three are $(M+1)\times(M+1)$ and tridiagonal; $\boldsymbol{A},\boldsymbol{B}$ symmetric, $\boldsymbol{C}^T=-\boldsymbol{C}$ *except* for its last diagonal entry, so $\boldsymbol{S}$ is symmetric and — since $V_h\subset V$ has no rigid-body mode left after clamping — positive definite.

Why the boxed entries. Exactly Lecture 6's argument for the Neumann end node ("the last diagonal entry is 1, not 2": every interior node has two elements attached, the end node only one). Writing the Hermite element matrix with $EI=1$, $\frac{1}{h^3}\begin{bmatrix}12 & 6h & -12 & 6h\\ 6h & 4h^2 & -6h & 2h^2\\ -12 & -6h & 12 & -6h\\ 6h & 2h^2 & -6h & 4h^2\end{bmatrix}$, an interior node collects $12+12=24$, $4h^2+4h^2=8h^2$ and $-6h+6h=0$; the tip collects only the left element's 12, $4h^2$ and $-6h$. That is $\boxed{1}$, $\boxed{2}$ and $\boxed{-1}$ in the scalings of (22).

Step 6 — the load vector [Lecture 12, eqs. (1.195)–(1.198)]

- By definition,

$$b_k^{(1)}=\int_0^1 f\,\phi_k^{(1)}\,\mathrm{d}x,\qquad b_k^{(2)}=\int_0^1 f\,\phi_k^{(2)}\,\mathrm{d}x,\qquad k=1,\dots,M+1. \tag{23}$$

- Lecture 12 evaluates these by interpolating the load with the same Hermite basis. For an *interior* node $k\le M$ that gives (1.197)–(1.198),

$$\begin{aligned} b_k^{(1)} &= \tfrac{h}{70}\big[9f_{k-1}+52f_k+9f_{k+1}\big]+\tfrac{13h^2}{420}\big[f'_{k-1}-f'_{k+1}\big],\\ b_k^{(2)} &= \tfrac{13h^2}{420}\big[f_{k+1}-f_{k-1}\big]-\tfrac{h^3}{420}\big[3f'_{k-1}-8f'_k+3f'_{k+1}\big]. \end{aligned}$$

- At the **tip** $k=M+1$ every term carrying x_{M+2} is absent, and only the left element contributes:

$$\begin{aligned} b_{M+1}^{(1)} &= \tfrac{h}{420}\big[54f_M+156f_{M+1}\big]+\tfrac{h^2}{420}\big[13f'_M-22f'_{M+1}\big],\\ b_{M+1}^{(2)} &= -\tfrac{h^2}{420}\big[13f_M+22f_{M+1}\big]-\tfrac{h^3}{420}\big[3f'_M-4f'_{M+1}\big]. \end{aligned}$$

- (In Problem 6 the integrals (23) are evaluated by Gauss quadrature instead, which avoids interpolating f at all.)

Mechanics reading. $\boldsymbol{\xi}$ are nodal deflections, $\boldsymbol{\eta}$ nodal rotations, $\boldsymbol{b}^{(1)}$ consistent nodal forces and $\boldsymbol{b}^{(2)}$ consistent nodal moments. (22) is the standard Euler–Bernoulli beam assembly with $EI=1$, sorted by DOF type rather than by node. The free tip keeps both DOFs and gets a half-stiffness row — which is just the statement that only one element frames into it.

Problem 4

Problem. Prove an *a priori* error estimate for this method in the energy norm

$$\|e\|_E = \Big\{ \int_0^1 (e'')^2 \,\mathrm{d}x \Big\}^{1/2}.$$

Throughout, $e := u - U$ (exact minus FEM). Two norms, told apart by the subscript: $\|v\|_E = \big(\int_0^1 (v'')^2 \,\mathrm{d}x\big)^{1/2}$ (energy, the one asked for) and $\|v\| = \big(\int_0^1 v^2 \,\mathrm{d}x\big)^{1/2}$ (L^2). Note $\|v\|_E = \|v''\|$: the energy norm is just the L^2 norm of the curvature. Lectures 6–7 carry a coefficient a in these; here $a \equiv 1$.

Step 1 — Galerkin orthogonality [Lecture 6, eq. (1.95): write both problems, subtract]

- The exact solution satisfies (5) against *every* $v \in V$:

$$\int_0^1 u'' v'' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \forall\, v \in V.$$

- The FEM solution satisfies the same identity, but only against $v \in V_h$:

$$\int_0^1 U'' v'' \,\mathrm{d}x = \int_0^1 f v \,\mathrm{d}x \qquad \forall\, v \in V_h.$$

- $V_h \subset V$, so the first line is also true for every $v \in V_h$. Subtract, for those v — the right-hand sides are identical and cancel:

$$\boxed{\int_0^1 e''\, v'' \,\mathrm{d}x = 0 \qquad \text{for all } v \in V_h, \qquad e = u - U} \tag{24}$$

Conforming ($V_h \subset V$) is what makes the subtraction legal: without it the exact solution's identity would not be available for the v we want to use.

Step 2 — Cauchy's inequality, in the form needed here [Lecture 6, eq. (1.91)]

- Lecture 6 states it as $\big|\int_0^1 a\, v' w' \,\mathrm{d}x\big| \le \|v\|_E \|w\|_E$.
- For this problem the pairing is of *second* derivatives with $a \equiv 1$, so the same inequality reads

$$\boxed{\Big|\int_0^1 v'' w'' \,\mathrm{d}x\Big| \le \|v\|_E \, \|w\|_E} \qquad \|v\|_E = \Big(\int_0^1 (v'')^2 \,\mathrm{d}x\Big)^{1/2}, \quad \|w\|_E = \Big(\int_0^1 (w'')^2 \,\mathrm{d}x\Big)^{1/2}. \tag{25}$$

Step 3 — U is the best approximation in the energy norm [Lecture 6, eqs. (1.96)–(1.100): insert $-v+v$, kill one term by (24), apply Cauchy, cancel one factor]

- Start from the definition, both factors e:

$$\|e\|_E^2 = \int_0^1 e''\, e'' \,\mathrm{d}x.$$

- Pick *any* $v \in V_h$ and insert $-v + v$ in the second factor, $e = u - U = (u - v) + (v - U)$:

$$\|e\|_E^2 = \int_0^1 e''(u - v)'' \,\mathrm{d}x + \underbrace{\int_0^1 e''(v - U)'' \,\mathrm{d}x}_{=\,0 \text{ by } (24)}. \tag{26}$$

- **Why the second integral vanishes.** (24) kills $\int e''(\,\cdot\,)''$ against anything *in* V_h. Here $v \in V_h$ and $U \in V_h$, so $v - U \in V_h$ and the integral is zero. Note $u - v$ is *not* in V_h (u is not a piecewise cubic) — that is exactly why the split puts $v - U$ in the slot being annihilated.
- So for any $v \in V_h$, $\|e\|_E^2 = \int_0^1 e''(u - v)'' \,\mathrm{d}x$. Apply (25):

$$\|e\|_E^2 \;\le\; \|e\|_E \, \|u - v\|_E.$$

- Cancel one factor $\|e\|_E$ (if it is zero, $U = u$ in the energy norm and there is nothing to prove):

$$\boxed{\|u - U\|_E \;\le\; \|u - v\|_E \qquad \text{for every } v \in V_h} \tag{27}$$

(27) says the FEM solution is the **closest** member of V_h to u in the energy norm — no other piecewise cubic on this mesh stores its bending energy more accurately. The constant is exactly 1. The error of the *method* is now bounded by the error of *interpolation*, which has nothing to do with the method and everything to do with the mesh.

Step 4 — choose v to be the interpolant [Lecture 12, eq. (1.182): $\Pi_h u$ matches u *and* u' at every node]

- (27) holds for *every* $v \in V_h$, so we are free to pick the most convenient one. Take the Hermite interpolant

$$\Pi_h u = \sum_{j=1}^{M+1} u(x_j)\, \phi_j^{(1)} + \sum_{j=1}^{M+1} u'(x_j)\, \phi_j^{(2)}, \tag{28}$$

  the piecewise cubic that agrees with u in *value and slope* at every node.
- **Check $\Pi_h u \in V_h$.** Every ϕ in (28) vanishes together with its slope at $x_0 = 0$, so $\Pi_h u(0) = (\Pi_h u)'(0) = 0$: the essential BCs hold, and $\Pi_h u$ is a legal choice. (Node 0 is absent from the sums, but $u(0) = u'(0) = 0$ anyway, so $\Pi_h u$ still interpolates u at *all* $M + 2$ nodes.)
- Put $v = \Pi_h u$ into (27):

$$\|u - U\|_E \;\le\; \|u - \Pi_h u\|_E \;=\; \|\big(u - \Pi_h u\big)''\|. \tag{29}$$

Step 5 — insert the interpolation estimate [Lecture 12, eq. (1.185): $\|(u - \Pi_h u)''\| \le c_3 \|h^2 u^{(4)}\|$]

- Lecture 12 states the three Hermite interpolation errors; the one that matches (29) is the second-derivative line,

$$\|\big(u - \Pi_h u\big)''\|_{L^2(0,1)} \;\le\; c_3 \, \|h^2 u^{(4)}\|_{L^2(0,1)}. \tag{30}$$

- Combining (29) and (30):

$$\boxed{\|u - U\|_E \;\le\; c_3 \, \|h^2 u^{(4)}\|} \tag{31}$$

- On a uniform mesh h is a constant and comes out of the norm:

$$\boxed{\|u - U\|_E \;\le\; c_3 \, h^2 \, \|u^{(4)}\| \;=\; \mathcal{O}(h^2)} \tag{32}$$

Result. For CG(3) on the cantilever, with $u^{(4)} = f$ square integrable,

$$\|u - U\|_E = \left\{ \int_0^1 \big((u-U)''\big)^2 \, \mathrm{d}x \right\}^{1/2} \;\le\; c_3 \, h^2 \, \|u^{(4)}\|,$$

i.e. the energy-norm error converges at **second order** in h. Halving the mesh divides the error by 4.

Reading the result. The energy norm measures the error in u'', the *curvature* — and curvature is bending moment ($M = EI\,u''$). So this is the convergence rate of the *moment/stress* field, $O(h^2)$. Problem 5 does the displacement, which will turn out two orders better. Note also $u^{(4)} = f$ by the PDE, so the bound can be written $c_3 h^2 \|f\|$: the rougher the load, the worse the constant, and if f is not square integrable the estimate says nothing.

Problem 5

Problem. Prove an *a priori* error estimate for this method in the L^2 norm

$$\|e\|_{L^2} = \Big\{ \int_0^1 (e)^2 \, \mathrm{d}x \Big\}^{1/2}.$$

Written $\|e\|$ below. The energy norm measured e''; this one measures e itself. The whole point is that it converges *faster*, and the device that proves it is the duality argument of Lectures 7–8.

Step 1 — the dual problem [Lecture 7, eq. (1.108): an auxiliary BVP whose *load is the error itself*]

- Lecture 7 solves $-(a\phi')' = e$ for the second-order problem. The beam operator is $\mathrm{d}^4/\mathrm{d}x^4$, so the same device reads

$$\boxed{\phi^{(iv)} = e \ \text{ on } (0,1), \qquad \phi(0) = \phi'(0) = 0, \qquad \phi''(1) = \phi'''(1) = 0} \tag{33}$$

  — the *same* cantilever BCs as the original problem.
- As in Lecture 7, this is legitimate because the problem is **self-adjoint**: the dual operator is the same $(\cdot)^{(iv)}$, so we are not obliged to solve a different equation.
- ϕ is never computed. It exists only to be paired against e, which is the next step.

Step 2 — turn $\|e\|^2$ into an integral of $e''\phi''$ [Lecture 7, eq. (1.109): multiply the dual equation by e, integrate by parts, boundary terms vanish]

- Multiply (33) by e and integrate — the load of the dual problem is e, so this manufactures $\|e\|^2$ on the left:

$$\|e\|^2 = \int_0^1 e\, e \, \mathrm{d}x = \int_0^1 e\, \phi^{(iv)} \, \mathrm{d}x.$$

- Integrate by parts twice, exactly as in Problem 1:

$$\int_0^1 e\, \phi^{(iv)} \, \mathrm{d}x = \underbrace{\Big[e\, \phi''' \Big]_0^1}_{\text{A}} - \underbrace{\Big[e'\, \phi'' \Big]_0^1}_{\text{B}} + \int_0^1 e''\, \phi'' \, \mathrm{d}x. \tag{34}$$

- All four boundary halves vanish — but for two *different* reasons:

term	dies because	which function supplies it
$e(1)\,\phi'''(1)$	$\phi'''(1) = 0$, a *natural* BC of (33)	the dual ϕ
$e'(1)\,\phi''(1)$	$\phi''(1) = 0$, a *natural* BC of (33)	the dual ϕ
$e(0)\,\phi'''(0)$	$e(0) = u(0) - U(0) = 0 - 0 = 0$	the error e
$e'(0)\,\phi''(0)$	$e'(0) = u'(0) - U'(0) = 0 - 0 = 0$	the error e

- (The $x = 0$ rows hold because *both* u and U lie in V, so each has zero value and zero slope at the clamped end; their difference does too.) Hence

$$\|e\|^2 = \int_0^1 e''\, \phi'' \, \mathrm{d}x. \tag{35}$$

Step 3 — subtract zero, using Galerkin orthogonality [Lecture 7, eq. (1.111): replace ϕ by the interpolation error $\phi - \Pi_h\phi$]

- $\Pi_h\phi \in V_h$ (same check as Problem 4: $\phi(0) = \phi'(0) = 0$), so (24) gives $\int_0^1 e''(\Pi_h\phi)'' \,\mathrm{d}x = 0$.
- Subtract that zero from (35):

$$\|e\|^2 = \int_0^1 e''\phi'' \,\mathrm{d}x - \underbrace{\int_0^1 e''(\Pi_h\phi)'' \,\mathrm{d}x}_{=0} = \int_0^1 e''\big(\phi - \Pi_h\phi\big)'' \,\mathrm{d}x. \tag{36}$$

This single move is the whole trick. Nothing was added or removed — the subtracted term is zero. But ϕ'' is not small, whereas the *interpolation error* $(\phi - \Pi_h\phi)''$ is: it is $O(h^2)$. That substitution is what buys the extra powers of h.

Step 4 — Cauchy, then interpolation, then the dual equation [Lecture 6 eq. (1.91) and Lecture 12 eq. (1.185), as in Problem 4]

- Cauchy (25) on (36):

$$\|e\|^2 \;\le\; \|e''\| \,\big\|\big(\phi - \Pi_h\phi\big)''\big\| \;=\; \|e\|_E \,\big\|\big(\phi - \Pi_h\phi\big)''\big\|.$$

- Lecture 12's interpolation estimate (30), now applied to ϕ instead of u:

$$\big\|\big(\phi - \Pi_h\phi\big)''\big\| \;\le\; c_3\, h^2\, \|\phi^{(4)}\|.$$

- **Here the dual problem pays off.** $\phi^{(iv)} = e$ by (33), so $\|\phi^{(4)}\| = \|e\|$ *exactly* — no stability constant is needed. (Lecture 7 needs S_a because of the coefficient a; its own note records that for $a \equiv$ const the ratio is explicit. Here the coefficient is 1 and the dual load *is* the error, so the constant is 1.)
- Therefore

$$\|e\|^2 \;\le\; c_3\, h^2\, \|e\|_E\, \|e\|,$$

  and cancelling one factor $\|e\|$ (if it is zero there is nothing to prove):

$$\boxed{\|e\| \;\le\; c_3\, h^2\, \|e\|_E} \tag{37}$$

Step 5 — insert Problem 4

- Problem 4 gave $\|e\|_E \le c_3 h^2 \|u^{(4)}\|$, (32). Substituting into (37):

$$\boxed{\|e\|_{L^2} \;\le\; c_3^2\, h^4\, \|u^{(4)}\| \;=\; \mathcal{O}(h^4)} \tag{38}$$

Result. For CG(3) on the cantilever,

$$\|u - U\|_E \le c_3 h^2 \|u^{(4)}\| = \mathcal{O}(h^2), \qquad \|u - U\|_{L^2} \le c_3^2 h^4 \|u^{(4)}\| = \mathcal{O}(h^4).$$

The L^2 error converges **two orders faster** than the energy-norm error.

Where the two extra powers came from. The interpolation estimate was used *twice*: once on e through Problem 4, and once on ϕ through (36). Each contributes h^2. This is the duality (Aubin–Nitsche) argument

of Lecture 7, one order higher because the beam's interpolation estimate carries h^2 where the bar's carried h.

Engineering reading. Displacements converge far faster than moments. $\|e\|_E$ measures curvature, i.e. bending moment: $O(h^2)$. $\|e\|_{L^2}$ measures deflection: $O(h^4)$. On a mesh where the deflection plot looks perfectly converged, the moment plot can still be visibly wrong — which is why stress, not displacement, drives mesh refinement. These are the two rates Problem 6 measures numerically.